

POKHARA UNIVERSITY

Level: Bachelor

Semester – Spring

Year : 2011

Programme: BE

Full Marks: 100

Course: Microprocessors

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. Briefly discuss the evolution of microprocessors. Also mention the important features of different generations. 5
- b. Explain the different addressing modes of Intel 8085 microprocessor. 5
- c. Differentiate between von-Neumann and Harvard architecture. 5
2. a. Sixteen bytes of data available from memory location 2500H to 2510H. Write an ALP for 8085 to count even and odd number and store the count in the register B and C for even and odd number respectively. 5
- b. Draw the timing diagrams for the instruction MVI 24H. 5
- c. Explain the function of following pins in 8085 microprocessor. SID, HLDA, RESET, READY and INTR 5

OR

- a. Draw and describe the timing diagram of I/O write cycle for 8085. 5
- b. Write an assembly language program using 8085 instructions to generate a Fibonacci series up to eight terms. 5
- c. WAP in 8085 to transfer ten bytes data from 5050H to 5060H only if data is between 30H and 70H else store 00H. 5
3. a. Write an 8086 Assembly Language program which will input the user name from the keyboard. If the user is "Pokhara University" it will give the output "valid user" else it will output "invalid user". 7
- b. Define assembler directives. Discuss the function of following directives. EQU, PROC, DOSSEG, DD, ENDP, ASSUME 8
4. a. WAP in 8086 to transfer 100bytes from 2000:0000H to output port 00H with a delay between each transfer. 7
- b. Explain with suitable examples the different addressing modes used in 8086? 8

5. a. Explain with a circuit diagram how 2048×8 ROM is interfaced with 8085 microprocessor. 8
- b. Differentiate between asynchronous and synchronous serial data transfer. Draw the accumulator bit pattern for synchronous data transfer. 7

OR

- a. With neat and clean diagram describe the working operation of SRAM and DRAM. 8
- b. Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 7
6. a. Explain hardware and software interrupts. Also explain the chained interrupts concept for handling multiple interrupts. 8
- b. Compare and contrast between RISC and CISC processor. 7
7. Write short notes on **any two**: 2×5
- a. Flags of 8085 microprocessor
- b. Interrupt vector table
- c. Automated calculator

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester – Fall

Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What features contrast the microprocessor with microcontroller? 10
Compare Von-Neumann architecture and Harvard architecture
- b) What are the greatest breakthroughs achieved in the successive development of microprocessor? List with their features. 5
2. a) Define addressing mode and explain all the addressing modes in 8085 with the individual figure and example associated with each 8
- b) Write the program in 8085 to count the number of positive, negative and zero in a given series of data. The length of series is given at address 203FH and the series starts from 2040H store your result at 2060H onwards 7
3. a) Draw the labeled timing diagram for the instruction MVI A, Data i.e. You can assume any data of 8 bit in length 8
- b) Write an 8086 ALP using MASM to accept a string from keyboard and display it in console in reverse order. 7
4. a) Specify the functions of the following 8086 instructions: LAHF, XLAT, IRET, IMUL, LEA, CWD and AAM. 7
- b) Write an assembly language in 8086 which transfer 10 byte of data stored in "array 1" to the new location "array2". 8
5. a) Write a program in 8086 which displays "Hello World" on the screen 7
- b) What are the different types of memory? Explain with specific application. 8
6. a) Explain the RS232 C standard for serial communication. 8
- b) Define interrupt. Explain the chained and polled concept of interrupt priority. 7
7. Write short notes on **any two**: 2×5
 - a) RISC and CISC
 - b) Flags of 8051
 - c) RTL

POKHARA UNIVERSITY

Level: Bachelor

Programme: BE

Course: Microprocessors

Semester: Spring

Year : 2012

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Define the terms microprocessor and microcontroller? What are the major enhancements on Harvard Architecture than Von Neumann architecture? Explain with necessary diagrams. 7
- b) Discuss various types of addressing modes of 8085 microprocessor with example. 8
- a) Draw and explain the timing diagram for the instruction STA C000H of 8085 microprocessor. 8
- b) Write an 8086 ALP using MASM to accept a string from keyboard and display it in console in reverse order. 7
- a) List some features of 8086 microprocessor. Describe its internal block diagram. How it is different from 8085? 7
- b) Write a program in 8085, sixteen bit data are stored in two tables starting at 4050H and 4070H, 10 data in each table. Add corresponding data and store it in third table starting at 4090H. 8
- a) Interface two RAM each having memory capacity 2Kx8 and two ROM each having memory capacity 4Kx8. The starting memory address is 8000H 8
- b) Explain in detail the RS-232C standard. 7
- a) What do you mean by interrupt? Explain about Maskable and Non-Maskable interrupt. 7
- b) Write a program to find factorial of a number in 8086 microprocessor. Write comment in each instruction. 8
- a) Discuss about RISC & CISC processors with examples. 8
- b) What do you mean by interrupt system? Illustrate the implementation.

details of polling and chained interrupts for handling multiple interrupts in 8085 microprocessor. 7

7. Write short notes on: (Any two)

- a) Flags of 8086.
- b) Synchronous versus Asynchronous communication.
- c) SIM & RIM.

2×5

7

8

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8

POKHARA UNIVERSITY

Level: Bachelor
Programme: B.E
Course: Microprocessors

Semester – Fall

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the evolution of Microprocessor. The processing speed of Harvard architecture is higher than Von Neumann machine, Explain Why? 8
b) Draw the functional block diagram of 8085 microprocessor and explain the functions of each block in detail. 7
2. a) Draw and explain the timing diagram for the instruction LDA 9000H 8
b) Write an ALP for 8085 to count the integers available in an array starting from memory location 2500H to 2510H for exactly divisible by two and save the result in the register B. 7
3. a) Explain the addressing modes of 8086 with suitable examples. 8
b) Write an 8086 ALP which will input the student's registration number from the keyboard. If the registration number is valid it will display the students name else it will display 'Not enrolled' 7
4. a) The six data bytes are stored from memory location 3000H to 3005H. Write an 8086 ALP to transfer block of data to new location 5000H to 5005H. 7
b) Draw an address decoding circuit for 8085 to interface 2KB RAM and 4KB ROM. 8
5. a) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standard? 7

- b) What is Interrupt priority of 8085 Microprocessor? Write an assembly language program to check to see if RST 6.5 is pending, assuming the microprocessor completing RST 7.5 interrupt request. If it is pending, enable RST 6.5 without affecting any other interrupts; otherwise, return to the main program.
6. a) Explain how the 20 bit physical address is calculated in 8086 Microprocessor? Compare 8086 Microprocessor with 8085 Microprocessor
- b) WAP in 8085 to transfer ten bytes data from 5050H to 5060H only if data is between 30H and 70H else store 00H in the next table.
7. Write short notes on **any two**:
- a) Macro assembler (8086)
 - b) Flags of 8086
 - c) RISC vs. CISC

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Harvard architecture and give reason behind this architecture being faster than Von-Neumann architecture.
b) Define flags and addressing modes. What are the different addressing modes used in 8085 microprocessor? Explain with examples.
2. a) Draw the timing diagrams for the instruction IN 4FH.
b) Write a 8085 program to find the largest of three numbers.
3. a) Write an 8085 ALP to find ODD and EVEN number among six bytes of data: 23H, 41H, 56H, AFH, C5H & A7H stored in the memory location starting from 2500H and place the ODD and EVEN numbers in memory location starting from 2600H and 2400H respectively.
b) Design a circuit to interface a 8 KB ROM to 8085 microprocessor. The starting address being 1000H.
4. a) Draw the functional block diagram of 8086 Microprocessor and explain how segmentation and pipelining concepts are implemented in this microprocessor.
b) Write an assembly language programming for 8086 to find the square root of a given number. Assume that a number is of two digits and is perfect square.
5. a) Specify the functions of the following 8086 instructions: AAA, LDS, DIV, MOVSB, NOT, CALL, ROL
b) Define IVT. Explain interrupt vector table for 8086 microprocessor.
6. a) Explain the RS232C standard for serial communication. Explain its advantages and disadvantages.
b) Explain the functional block diagram of 8259 PIC.
7. Write short notes on: (Any two)
a) Synchronous and Asynchronous data bus
b) Modes of operation of 8255
c) Memory mapped I/O and I/O mapped I/O.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Fall

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the Von Neumann's architecture of a computer system. 5
b) Differentiate between a microprocessor and microcontroller. 5
c) Explain the addressing modes of Intel 8085. 5
2. a) Starting from the address $DOOO_h$ to $DOO9_h$ there is an array of ten data. Write a programme using 8085 instructions to find out how many times the data $08h$ repeats in the given array. Store the result at address $C000_H$. 7
b) Write an assembly language program for Intel 8085 to compute 8
$$\sum_{i=1}^n \frac{x_i}{2}$$

Where X_i are three numbers stored at memory locations $A001H$, $A002H$ and $A003H$. Store the result at memory location starting at $D000H$.
3. a) What is the significance of Timing Diagram? Draw the labeled timing diagram for the instruction $OUT\ 80H$. 8
b) Draw the block diagram of Intel 8255 PPI, and explain the functions of the sub blocks. 7
4. a) Explain how two 2048×8 RAM are interfaced with 8085 microprocessor and also find the address range for each RAM chips. 7
b) Write an 8086 ALP for MASM to find a square of a given number. 8
5. a) Draw the functional block diagram of 8086 microprocessor and compare the functions of its two basic units. 7
b) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 8

microprocessor and explain different types of 8086 interrupts with respect to interrupt vector table.

6. a) Write an 8086 ALP for MASM to display the string "POKHARA UNIVERSITY" without using 09h on screen. Explain all steps and assume necessary data. 7
- b) Two computers are to be connected with each other via modems. Using RS-232 standards, explain the connections? Also, show the connections for null modem. 8
7. Write short notes on: (Any two) 2×5
- a) Features of Intel 80386
- b) Synchronous & asynchronous data transmission
- c) Macros.

Level: Bachelor
 Programme: BE
 Course: Microprocessors

Semester: Spring

Year : 2014
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the essential differences between: 8
 - i. Von Neumann and Harvard Architecture.
 - ii. Microprocessor and Microcontroller.
- b) Define flags and addressing modes. Explain the role of all the flag bits in 8085 microprocessor with essential examples. 7
2. a) Explain the function of the following **8085** program. What will be the output of the program displayed at port **01H** when the program is executed if the value of **BYTE1** is equal to **FFH**. 5

```

                MVI A,BYTE1
                ORA A
                JM  OUTPRT
                OUT 01H
                HLT

OUTPRT:        CMA
                ADI 01H
                OUT 01H

                HLT
    
```

- b) Specify the function of the following instructions from **8085** instruction set. 5
 - i. LXI
 - ii. LDAX
 - iii. JC
 - iv. PUSH
 - v. SBB
- c) Write a 8085 program to sort given 10 numbers from memory location 5

2200H in the ascending order.

3. a) What is the significance of Timing Diagram? Draw the labeled timing diagram for the instruction OUT 4FH.
b) Explain how 1KB RAM, 2KB ROM and 4KB EPROM are interfaced with 8085 microprocessor and also find the address range for each chips.
4. a) Explain control word format of 8255 PPI for I/O mode.
b) Draw the functional block diagram of 8086 Microprocessor and explain it and illustrate how its architecture is faster than its predecessors.
5. a) Write an 8086 ALP for MASM to find the square of a given number.
b) Write a program to reverse the given string of 8086.
6. a) Define "Assembler Directives". Explain the following assembler directives along with their format: The Memory Model Definition, The PROC directive, The .STARTUP directive, The .EXIT directive and The END directive.
b) Two computers are to be connected with each other via modems Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards?
7. Write short notes on: (**Any two**)
 - a) One pass and two pass assembler.
 - b) RS 422A.
 - c) Direct Memory Access.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

7.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define microprocessor. What are the essential differences between Von Neumann and Harvard Architecture? 5
b) Explain in detail the addressing mode of 8085 microprocessor. 5
c) Write an 8085 ALP to find out the largest number in an array available from memory location starting from 2500H to 2510H and Store the result in the register B. 5
2. a) Write a Program to check 1's complement of the data stored in address C000H and store the result in D000H. 7
b) Draw the timing diagrams for the instruction LDA 400FH. 8
3. a) Design a memory interface circuit to interface two chips of 4K and another chip of 2K ROM of 8085 without fold back addresses, starting address is 8000H. 7
b) Define DMA. Explain mechanism of DMA with block diagram. 5
c) Define "Assembler Directives". Discuss the function of following directives: SEGMENT AND GLOBAL. 3
4. a) Draw and explain the functional block diagram of 8255A PPI. 7
b) Write an 8086 ALP in masm to display the string "POKHARA UNIVERSITY" in reverse order. 8
5. a) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor and explain different types of 8086 interrupts with respect to interrupt vector table. 8
b) Write an 8086 ALP to find the factorial of 08H using DOS BIOS Interrupt. 7
6. a) Two computers are to be connected with each other via modems Using RS-232 standards, explain the connection. How can they be 7

connected directly without any modems using RS 232 standards?

b) Explain in detail the flags of 8086 microprocessor.

7. Write short notes on: (**Any two**)

a) One pass and cross assembler.

b) RS 422A.

c) Bus structure.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Enlist the greatest breakthrough in microprocessor so that modern processors are available for personal computer. 5
b) With reference of bus advancement differentiate between Harvard Architecture and Von Neumann architecture. 5
c) Explain with example the addressing mode of 8085 microprocessor. 5
2. a) Define instruction set and formats, opcodes, mnemonics and operands with examples. 8
b) Write an 8085 program for the following type of addition 7
$$1^2 + 2^2 + 3^2 + \dots + 9^2$$
3. a) Define T-state. Draw the timing diagrams for the instruction OUT 0FH. 8
b) Write a program to take input from 4- switches connected to PC3-PC0 and display the status of the switches to 4 LEDs connected to PC7 – PC4 of 8255 PPI. 7
4. a) What is memory interfacing? Interface two PROM and one RAM with 8KB and 4KB memory respectively with 8085. 8
b) Explain the Programmable Interrupt Controller (8259 A) with a suitable block diagram. 7
5. a) Do you think micro assembler is essential for microprocessor to execute the program? If yes, explain with the difference between 7
 - i. END and ENDP
 - ii. DB and DW
 - iii. PROC and EXTRN
 - iv. PUBLIC and GLOBAL

- b) Write an 8086 program to find square root of a given number. Given that number is a perfect square of two digits.
6. a) Explain with example the flags of 8086 microprocessor.
b) Write the difference between serial and parallel communication. Explain the RS232C standard for serial communication.
7. Write short notes on: (**Any two**)
a) 8255 PPI
b) 80386 microprocessor and its type
c) Procedures and Macros

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Microprocessor. Compare Intel 8085, 8086 and 80386 Microprocessors on the basis of their features and internal architectures. 7
b) Explain flags in 8085 with the help of a diagram. What is the function of LDAXB? 8
2. a) Write an ALP in 8085 to find whether the given number is palindrome or not. 8
b) Write a 8085 program to find smallest of three numbers. 7
3. a) Define T-state. Draw the labeled timing diagram for the instruction LXI B, C00FH. 7
b) What are the three basic functions microprocessor should satisfy to interface any memory chip? Explain how 2048×8 EPROM is interfaced with 8085 microprocessor. 8
4. a) What are interrupts? Differentiate between I/O mapped I/O and Memory mapped I/O. 7
b) Draw and explain the functional block diagram of 8259A PIC. 8
5. a) Write an assembly language program for 8086 to find the sum of two numbers input by the user through the keyboard and display the sum on the screen. 8
b) Write an 8086 ALP for MASM in DOS-BIOS mode to display the string "Pokhara University" on screen without using 09H. 7
6. a) Explain procedures and macros. Why IVT is used in microprocessor. 7
b) Draw and explain the connections to show how RS-232C is made compatible with TTL logic along with its connections for null modem. 8
7. Write short notes on: (**Any two**) 2×5
 - a) HOLD and HLDA pin
 - b) BSYNCR Protocol
 - c) DMA

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Von-Neumann and Harvard architecture. 5
b) Explain in detail the flags of 8085 microprocessor. 5
c) Write an 8085 ALP to find out the smallest number in an array available from memory location starting from 2500H to 2510H and Store the result in the register C. 5
2. a) What happens when HLT instruction is executed in processor? 4
b) Write an 8085 ALP to multiply the number 37H and 42H and keep the result in memory. 5
c) Design a circuit to interface a 2 KB EPROM to 8085 microprocessor .The starting address being 1000H. 6
3. a) Draw the timing diagrams for the instruction STA 400FH. What does quality factor mean in microprocessor? 7+1
b) What is control word? Write a program to take input from 8 switches connected to port B of 8255A and display the status of switches of 8, LED connected to port A of 8255A (for mode 0). 7
a) With the help of the register organization diagram explain the types of registers in 8086 microprocessor. 7
b) What is interrupt and interrupt service routine? Explain polled, vectored interrupt and non-maskable interrupt with examples. 8
5. a) Define "Assembler Directives". Explain the following assembler directives along with their format: The Memory Model Definition, The PROC directive, The .STARTUP directive, The .EXIT directive and The END directive. 7
b) Define Procedure and Macros and explain the addressing mode used in 8086 microprocessor with examples. 8

5. a) Write an 8086 ALP for MASM to display the text "POKHARAUNIVERSITY" in reverse order. 8
- b) Two computers are to be connected with each other via modems Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 7
7. Write short notes on: (Any two) 2×5
- a) RS 422A
 - b) SRAM Vs DRAM
 - c) 80386 microprocessor and its type

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare and contrast Microprocessor and Microcontroller. 7
b) Draw the block diagram for the internal architecture of Intel 8085 and explain in brief. 8
2. a) Explain in brief the different addressing modes of 8085 with proper examples. 7
b) Write an 8085 ALP that checks an array of ten memory location starting from C000H for odd numbers including zero and transfers the odd numbers in memory location starting from C00EH. 8
3. a) Explain bus structure of a Microprocessor. What do you mean by Synchronous and Asynchronous Bus? 7
b) With a suitable diagram explain the read and write operation with a SRAM cell. 8
4. a) Draw the block diagram of 8255 PPI and explain in brief. 7
b) What are the significance of assembler's directive in microprocessor? List all the assembler's directive of 8086 microprocessor and explain them with example. 8
5. a) How interrupt processing occurs in a microprocessor? Explain vector chain and polled interrupt. 7
b) Write an assembly language program in 8086 which reads a string from a keyboard and then displays the string in the reverse order. 8
6. a) Write a program in 8086 to find whether a number is positive or negative. 7
b) RS232 forms a basis for serial data communication standard. Explain with diagram. 8
7. Write short notes on: (Any two) 2×5
 - a) Memory mapped I/O versus I/O mapped I/O
 - b) Flags in 8085
 - c) Serial Communication

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the Microprocessor based System with basic block diagram. 5
b) Why the speed of Von-Neumann architecture is low than Harvard architecture? Explain Von-Neumann architecture with various registers used in it. 5
c) Why the flag is also called PSW? Explain with the help of 8085 flag register. 5
2. a) What are the different rotate instructions used in 8085 instruction set? Explain with the help of a simple example for each of them. 7
b) What is the difference between a counter and a time delay loop? Explain with the help of a flowchart. 8
3. a) Classify the Bus on the basis of their function. Differentiate between Synchronous and Asynchronous Bus. 7
b) What do you mean by address decoding? Design a scalable address decoding circuit to interface 64-byte RAM memory and 128-byte ROM memory. 8
4. a) With a suitable diagram explain the write operation with a DRAM cell. 7
b) Write an assembly language program in 8086 which reads a string from the keyboard and then displays the string in the reverse order. For example if the input string read from the keyboard is "POKHARA UNIVERSITY", the output should be "YTISREVINU ARAHKOP". 8
5. a) Define addressing mode. Explain the addressing modes of 8086 with proper examples. 7
b) What is the significance of interrupt in microprocessor? Explain the 8

IVT with suitable figure.

6. a) Explain EU and BIU with the help of well labelled block diagram of 8086 microprocessors. 7
- b) Differentiate between synchronous and asynchronous serial data communication. Also explain briefly the protocols of synchronous serial data communication. 8
7. Write short notes on: (Any two) 2x5
- a) RTL Instruction Descriptions
- b) Memory mapped I/O Vs I/O mapped I/O
- c) Serial data transmission standards

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Von-Neumann and Harvard Architecture. Explain fetch, decode and execute cycle in microprocessor with block diagram. 8
- b) Draw the functional architecture of a 8-bit microprocessor along with diagram and briefly describe about the interrupts available in it. 7
2. a) Draw and explain a well labelled timing diagram of the instruction MVI A, 32H and calculate the total execution time if clock frequency is 2 MHz. 7
- b) Write 8085 assembly programming language to find the smallest number among ten 8-bit data's stored in memory location 5000H to 5009H. Also store that value in 9100H and display on output device 51H. 8
3. a) Define T-state, Machine cycle and Instruction cycle. What are the different addressing models address in 8085 microprocessor? 7
- b) What are the three basic functions microprocessor should satisfy to interface any memory chip? Explain how 4096*8 PROM is interfaced with 8085 microprocessor. 8
4. a) Draw and explain the control word for 8255A PPI. Also, explain the different operating modes of 8255 PPI. 7
- b) Write an assembly language program in 8086 to display the string "Computer Engineering" at console. 8
5. a) What do you understand by macros and procedures? State and explain the different types of assembler directives in 8086 8
- b) What are the different dedicated interrupt pointers in 8086 microprocessor? Draw and explain the block diagram of 8086 IVT 7

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2018

Programme: BE

Full Marks: 100

Course: Microprocessor Systems and Interfacing

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by address decoding? Define and differentiate I/O mapped I/O and Memory mapped I/O.
b) Draw the functional block diagram of 8051 microcontroller describe each block briefly.
2. a) Define address decoding. Design a circuit to interface an 8 KB ROM, a 4 KB RAM and a 2 KB EPROM with a microprocessor.
b) A switch is connected to INTO pin and a LED is connected to P1.0 of 8051 microcontrollers. The INTO receives a logic high when switch is not pressed and logic low when switch is pressed. Write a program and an interrupt service routine to turn on the LED for 2 seconds if switch is pressed. Assume your own clock frequency for the system.
3. a) Draw the block diagram of 8253 PIT and explain each block in detail.
b) Write a program for 8255 to set bit PC3 and PC2 in BSR mode and RESET them after 10 seconds .The address of control register is 83H.
4. a) Define IVT and explain different types of interrupt in 8086. Explain any one application of hardware interrupt.
b) Explain the printer port connection for parallel communication.
5. a) Explain the terms reliability, fault tolerance, crosstalk and impedance matching.
b) What are the limitations of 8-bit ISA bus? Explain the advantages of using PCI bus.
6. a) Describe different noise coupling mechanism. Explain how shielding minimizes different types of noise coupling.

b) What are the types of shielding mechanism available? Explain with suitable diagrams. 7

7. Write short notes on: (Any two) 2×5

- a) PCB design
- b) PIC functional architecture
- c) Addressing modes of 8051

table along with related memory address.

6.
 - a) How DTE and DCE is interfaced with minimum lines using RS232 cable? Explain how two computers can be connected without modem.
 - b) What are the different methods of parallel data communication, Explain in detail.
7. Write short notes on: (Any two)
 - a) 8251- USART
 - b) Maximum mode and minimum mode of 8086
 - c) Microprocessor & Microcontroller

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the major differences between Microprocessor and Microcontroller? Why Harvard architecture is superior to Von-Neumann Architecture? 4+4
b) What are the interrupt pins of 8085 microprocessor? Explain the different methods of handling multiple interrupts 7
2. a) Write an assembly language program in 8085 to check whether the number stored in memory location D001 H is positive or negative. If positive store it in C001 else store in C002. Comments in program indicate full marks. 8
b) What are the different types of machine cycles in 8085 instructions? Draw the labelled timing diagram for the instruction LDA 30DA H. 2+5
3. a) Design an Addressing Decoding circuit to interface 2kB RAM, chips each of size 256 bytes with starting address 1001 H. 8
b) Define interrupts. Differentiate between I/O mapped I/O and Memory mapped I/O 7
4. a) Explain memory interfacing? How do you interface two chips of 4K RAM and one chip of 2K ROM of 8085 with and without fold back addresses, starting address is 8000H? 8
b) Write an 8086 ALP to display the string "POKHARA UNIVERSITY" in reverse order. 7
5. a) What are the different dedicated interrupt pointers in 8086 microprocessor? Draw and explain the block diagram of 8086 IVT table along with related memory address. 8
b) What is Macro assembler? Differentiate between Macros and Procedure. 7

6. a) What are the significance of assemblers directive in microprocessor? 8
List all the assembler's directive of 8086 microprocessor and explain them with example.
- b) What is Synchronous and Asynchronous data transmission? Explain the RS232C standard for serial communication. 2+5
7. Write short notes on: (Any two) 2×5
- a) SRAM and DRAM
 - b) RIM and SIM Instruction
 - c) RS 423A

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Von Neumann and Harvard architecture. Explain fetch, decode and execute cycle in microprocessor with block diagram. 8
- b) What is the importance of DAA instruction? Explain it with an appropriate example along with the necessary calculations. 7
2. a) Write an ALP for 8085 to count the integers available in an array starting from 7 memory location 3500H to 3510H for exactly divisible by two and save the result in the register B. 7
- b) What is the significance of timing diagram? Draw and Explain a well labelled timing diagram of the instruction XRA B and calculate the total execution time if clock frequency is 2 mhz. 8
3. a) What is the use of 8255 PPI? Draw and Explain the control word format of 8255 PPI in I/O mode. 8
- b) Define interrupts. Differentiate between I/O mapped I/O and Memory mapped I/O. 7
- a) Draw a circuit for 8085 to interface 2KB RAM and 4KB ROM 8
- b) What is Interrupt Vector Table (IVT)? Draw the IVT for 8086 microprocessor and explain different types of 8086 interrupts with respect to IVT. 7
5. a) Discuss the advantage of 8086 over 8085 and explain the concept of pipelining and segmentation 7
- b) Write an assembly language program in 8086 to find the largest number among 10 blocks of data and store the largest value in location "largest" 8

6. a) What is Macro assembler? Differentiate between Macros and Procedure. 7
- b) What do you understand by Null Modem? Show the interface of DTE and DCE with RS-232C. 8
7. Write short notes on: (Any two) 2×5
- a) DAM
 - b) Flags of 8085
 - c) Op codes, mnemonics and operands with example.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) What is microprocessor? Explain the evolution of microprocessors. 7
- b) Draw the internal architecture of 8085 microprocessor. Also, explain each blocks in brief. 8
2. a) What is addressing mode? Explain different addressing modes of 8085 with examples. 8
- b) Define machine cycle and instruction cycle. Draw the neat and labeled timing diagram for 8085 instruction NOP. 7
3. a) Write an ALP in 8085 to check whether the number stored in memory location 2060H is prime or not. If the number is prime, store FFH in memory location C00FH else store 00H. 8
- b) Define tri-state buffer. Explain with suitable circuit diagram how read and write operations are performed in SRAM chip. 7
4. a) Design an address decoding circuit to interface one ROM chip of 2KB and one RAM chip of 4KB at address 0000H and E000H respectively. 8
- b) Draw the suitable block diagram of 8255A PPI and explain. 7
5. a) What are assembler directives? Explain any six different assembler directives of 8086 microprocessor. 8
- b) Define IVT. Explain different pre-defined interrupts in 8086 microprocessor along with suitable block diagram. 7
6. a) Write an 8086 program to enter a string from the keyboard. Count the number of repetitions of letter 'a' or 'A'. If the count is even, display "POKHARA" else display "UNIVERSITY". 7
- b) Define DTE and DCE in serial communication. Explain RS232C serial data standard along with suitable circuit diagram. 8

7. Write short notes on: (Any two)

- a) Operation of CALL and RET instructions
- b) Peripheral- mapped I/O and Memory-mapped I/O
- c) Synchronous Vs asynchronous serial data communication

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Microcontroller. What are the essential difference between Von Neumann and Harvard architecture? 7
- b) Explain the architecture of 8085 microprocessor? 8
2. a) Draw the timing diagrams for the instruction STA E040H. The address and opcode of the instruction is shown below: 8

Address	Mnemonics	Op code
30FF	STA E040H	32H
3100		40H
3101		41H

- b) What are the types of instruction depending upon word size? Explain different type of addressing modes of 8085 microprocessor. 7
3. a) Draw and explain the functional block diagram of 8259 - PIC. 8
- b) What do you mean by unique and non-unique address decoding? Differentiate between synchronous and asynchronous bus. 7
4. a) Write an 8086 ALP to check whether a given string is palindrome or not. 7
- b) What do you understand by memory interfacing? Design an interfacing circuit to interface one 4KB EPROM and two 2KB RAM for 8085 microprocessor using 3*8 decoder. 8
5. a) What is interrupt vector table? Explain procedures and macros. 7
- b) Compare RS232 and RS422. Explain how two computers can communicate with each other using RS232 standard. 8

6. a) What is the Asynchronous serial data communication? Compare serial and parallel communication. Draw the Block diagram of Intel 8251A USART and explain functionally in detail. 8
- b) Write the RTL for the instruction LXI D, 9050H in 8085 microprocessor. Explain the operations and uses of DAA, SHLD and LHLD instructions in 8085. 7
7. Write short notes on: (**Any two**) 2×5
- a) Memory mapped I/O and I/O mapped I/O
 - b) Handshaking signals
 - c) Types of interrupt.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Microprocessor along with its applications. Differentiate Harvard and Von-Neumann architecture on the basis of storage. 7
b) What are the types of instruction depending upon word size? Explain different type of addressing modes of 8085 microprocessor. 8
2. a) Write an 8085 program to find smallest number among array of numbers. 8
b) Define T-state. Draw the labelled timing diagram of the instruction LXI H, 5051H. 7
3. a) Design an Addressing Decoding circuit to interface 16KB ROM chips with 8085 microprocessors. 8
b) Draw a well labelled architecture of 8086 microprocessor and discuss about BIU and EU. 7
4. a) Explain about IVT of 8086. 7
b) What are serial and parallel communication? Explain basics of RS-232. 8
5. a) Explain the working of DMA. Mention its advantages over programmed and interrupt driven I/O. 8
b) Write an ALP in 8086 to display POKHARA UNIVERSITY without using 09h function. 7
6. a) What is the Asynchronous serial data communication? Compare serial and parallel communication. Draw the Block diagram of 8251A USART and explain functionally in detail. 8
b) What is the importance of DAA instruction? Explain it with an appropriate example along with the necessary calculations. 7

7. Write short notes on: (Any two)

- a) Memory mapped I/O vs I/O mapped I/O
- b) Address Bus ,Data Bus and Control Bus
- c) Flags in 8086